



Hacking the predictive mind

Twenty years ago, in October 2003, I was a naive undergraduate student sat in the lecture theatre of the old Experimental Psychology building at the University of Oxford. The lecture was on the basics of vision. We were introduced to the classics of the field: Hubel and Wiesel, simple and complex cells, extrastriate regions processing colour, form and motion. Abstracting away from the details, the story told was of the ‘standard model’ of vision science—one in which sensory input arriving at the retina enters into a complex, feedforward processing chain, culminating in subjective perception.

I was unaware of it then, but this received wisdom was about to be turned on its head—slowly at first, and more recently in a flood of books and papers on ‘predictive processing’. From this revolutionary perspective, the brain is not a passive receiver of sensory information, but instead actively sculpts our experience of the world. Like all good ideas, this one has roots that can be traced a long way back, through the psychophysics of Hermann von Helmholtz in the 19th century, the mathematics of Thomas Bayes in the 18th, and the optics of Ibn al-Haytham in the 11th. In the modern era of cognitive science, Richard Gregory, in his influential 1970 book *The Intelligent Eye*, was an early proponent of the idea that perception is an active process of hypothesis-testing.¹ His ‘hollow face’ illusion is still one of the most compelling demonstrations of how top-down influences shape conscious experience. A face mask is rotated away from the observer, who continues to see a normal convex face in the concave side of the mask. For most people, it is impossible to see it any other way (Fig. 1). But in cognitive neuroscience, and allied branches of neurology and psychiatry, the feedforward, input-output computer metaphor of how the brain works held sway, particularly in the latter half of the 20th century. As the philosopher and cognitive scientist Andy Clark describes in *The Experience Machine*, only recently have these fields begun to re-discover and assimilate the radical perspective offered by predictive processing accounts of mind.²

Predictive processing holds that our experience of the world is continually being shaped by our background beliefs and expectations. Perception is never a passive process on this account, and instead the bulk of what the brain is doing is learning and maintaining a model of the world that is then used, moment to moment, to try to predict incoming sensory signals. When the internal model and the sensory input fail to match, a ‘prediction error’ ensues, and our hypotheses are subtly altered to better approximate reality.³ Such predictions are thought to be carried by backward connections in the brain (empirically estimated to substantially exceed the number of feedforward connections)—ones that flow outwards, towards the sensory periphery. Another feature of predictive processing architectures is that they are hierarchical—upper levels of the hierarchy both receive error signals arising from

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lower levels and send expectations back down. If you arrive at a friend’s house and know that they own a labrador, a higher-level expectation—the spatially invariant concept of ‘dog’—may be used to rapidly interpret and adjust your perception of the dog-shaped blurs that rush towards you as you open the door. A benefit of this architecture is that it is efficient, transmitting upwards only what cannot already be predicted based on an existing model of the world—an insight that has echoes in many algorithms used in modern computer vision.

This is the predictive processing story about perception. But *The Experience Machine* also covers a deeper (and perhaps less well-known) idea, active inference, championed by my UCL colleague (and Clark’s collaborator) Karl Friston.⁴ Just as predictive processing turns the standard model of perception on its head, active inference seeks to invert the conventional view of motor control as involving efferent commands. The idea is that perceptual and motor systems are both seeking to reduce prediction error, but via different mechanisms. Perception results from changing our hypotheses about the world to fit the sensory data. Action results from changing the sensory data to fit our hypotheses about the world. For instance, by thinking to ourselves, ‘I want to look over there’, we create a cascade of prediction errors due to our current direction of gaze being at odds with this expectation, such that resolving these error signals brings our gaze in line with what we intended. This is a profound—and to some, controversial—inversion of the conventional model of motor control. But it implies there may be a deep unity between the computational goals that drive both perception and action.

Much has been written in the past few years about predictive processing and active inference, including book-length summaries



Figure 1 Different angles of Phra Lak Lae, an example of the hollow face illusion in the face of the Buddha in Wat Hiranyawat temple, Thailand. [Chainwit., CC BY-SA 4.0 <<https://creativecommons.org/licenses/by-sa/4.0/>>, via Wikimedia Commons].

aimed at both technical and general audiences.^{4–6} Neuroscientific hypotheses about the implementation of predictive processing have been developed in detail, for instance under the guise of predictive coding.³ Clark himself has tackled the mapping between subjective experience and predictive processing accounts of mind, in his previous book *Surfing Uncertainty*.⁵ One thing that stands out about *The Experience Machine* is a focus on the practical consequences of understanding predictive processing—both for understanding clinical disorders, and also for optimizing our own lives. There is a lot for a clinician-scientist to grasp hold of here.

A central message is that we are not just passive sensors of what our bodies are doing, and instead medical symptoms of all kinds are active constructions. Clark marshals predictive processing in accounting for symptoms of functional neurological disorders, autism spectrum disorder, schizophrenia, post-traumatic stress

disorder and chronic pain. It turns out that in predictive processing accounts of clinical disorders, changes in precision-weighting—essentially, whether top-down or bottom-up factors come to dominate one's inference—do a lot of the work. Weighting signals by their precision (the inverse of their variance) makes logical sense, as only reliable predictions and prediction errors should contribute to our model of the world: in Clark's memorable phrase, precision estimates are the brain's way of 'telling itself where to place its bets'. But there is an intriguing regress here—precision itself needs to be estimated within the model, and if these estimates are awry, the system will behave 'as if' it has more or less precise data that it actually does.⁷ False expectations can in this way become self-reinforcing. For instance, in the case of functional disorders, a highly precise expectation that an affected body part or cognitive process is impaired may become a self-fulfilling prophecy. Such a framework can explain why distraction (the loosening of precision,

within predictive processing accounts) can make functional symptoms temporarily vanish, as in Hoover's sign. Clark deftly describes how the self-confirming nature of some priors can deepen our disconnection from reality—especially when those priors are about the body, or internal symptoms, and cannot easily be corrected by 'external' evidence.

The notion of relative balance is key here—and *The Experience Machine* is an up-to-date guide on how disturbances in such a balance may account for autism spectrum disorders on the one hand (as an overweighting of sensory evidence) and schizophrenia on the other. The latter story is subtle and still developing (see Corlett *et al.*⁸), but one influential hypothesis is that schizophrenia is defined by an excess of high precision (trusted) prediction errors, leading to alternative (delusional) models of reality being adopted in order to make sense of the world.

Towards the end of the book, Clark seeks to draw out practical lessons from the predictive processing framework under the rubric of 'hacking the prediction machine'. He describes how, because much of the predictive machinery operates outside of conscious awareness, 'honest' placebo effects—where the patient is aware that what they are receiving is a placebo—may still be as powerful as regular placebos. These medications may activate 'subterranean expectations through superficial indicators of reliability and efficacy'. More generally, this perspective suggests that pain is a delicate construction, and that 'pain receptors ... are only one source of evidence, acting in concert with knowledge and belief'. The powers of psychedelics and meditation are explored as potentially acting to loosen priors at the top of the hierarchy, and thus facilitate a literal 'change of mind'.

For much of the book, consciousness lurks as an unexplained ghost in the predictive machinery—leading to the smuggling in of qualifiers such as 'unconsciously expect', 'strong nonconscious predictions' and 'consciously or otherwise'.

But then, in a brief interlude halfway through, the heavy machinery of predictive processing is turned on consciousness itself. Clark is sympathetic to illusionism (though dislikes the term), suggesting that the apparent puzzle of consciousness is a 'trick of the mind', and that a successful predictive processing account will deflate the hard problem of consciousness by targeting the meta-problem.⁹ In Clark's sketch, consciousness has its roots in the human ability to meta-represent our own predictions: 'a major part of what we are predicting is ourselves'. Under this scheme, conscious agents do not have access to the 'processing' aspects of predictive processing (which remain, in Helmholtz's famous phrase, unconscious inferences), but we may have a capacity to meta-represent (all within a predictive model!) the beliefs, behaviours and responses that result. This kind of higher-order predictive model may then (i) only be conscious of certain things (not of its own workings, for instance); and (ii) be puzzled about why and how it knows things about the world and itself.

I am sympathetic to this approach but found myself wondering why certain things make it into this meta-model, and others don't. Consider this passage, about how predictive processing underpins the perception of a bird on a tree: 'Of course, you don't first see a rough tree outline, then a better one, this time with a bird added. It's all happening too fast for that. But there's a sense in which that's exactly what's going on in your brain.' (p. 30).

This sounds right, psychologically speaking. But why don't we see the dynamics of inference? Why is consciousness sluggishly following on, after the fact? Perhaps an account of the functions

of consciousness is required for us to know where to draw the dividing line between those aspects of the model that are experienced, and those that are not.¹⁰ In any case, this interlude was very thought-provoking but all too brief—perhaps the topic for a next book?

Clark writes beautifully, weaving in historical asides with a description of the theory that is bracingly clear and accessible, without requiring a background in psychology, neuroscience or mathematics. The science is interlaced with personal stories (being nervous before a big talk; exploring the West Yorkshire canals on a houseboat) that make you feel like you are in the company of an old friend. He is an astute guide to how predictive processing offers a psychologically productive and humane account of the mind. For instance, it elegantly explains why otherwise kind, well-intentioned, and intelligent human beings can see the same situation so differently. It offers a reminder of the depths of shared experience—no matter our background, race or religion, we are all biological agents, so when our deep-rooted expectations about our lives are disrupted, we feel the same levels of anguish and suffering. It also weakens the traditional divide between psychiatry and neurology. Under a predictive processing perspective, experiences such as functional symptoms are as 'organic' and real as any others, having their origins in a disturbance of the delicate balance of neural machinery maintaining our model of the world and of ourselves.

I personally feel the pull of this worldview—indeed, whenever I am thinking idly about a new research direction, or discussing an experiment with a student, my background assumptions about how things will unfold are often informed by an unarticulated toy model steeped in the ideas of predictive processing. But I also have a nagging doubt that predictive processing is sometimes too satisfying. Clark relates the story of pioneering Dutch microscopist Antonie van Leeuwenhoek who first observed spermatozoa under the microscope, in the 17th century. Van Leeuwenhoek was an advocate of preformationism—the belief that sperm contained 'preformed' adult humans—and he reported seeing nerves and arteries within his specimens. As Clark surmises, 'looking back on this today, we may well suspect that ... Leeuwenhoek's visual experience reflected nothing so much as those strong prior beliefs.' The irony is not lost on the reader. The predictive processing story is so powerful and all-encompassing that it is increasingly tempting to interpret new behavioural and neural data through its lens, potentially subtly affecting our credence in predictive processing itself.

We have been through revolutions before, and as Kuhn tells us, paradigms shift. It would be surprising if predictive processing is our last, even though it often feels like the right and proper way for thinking about how the mind works. Even within computational cognitive neuroscience the sea-change has been uneven—it has influenced how many researchers think about perception and action (and to some extent, affect and metacognition)—but has had less impact, so far, on the fields of learning, memory and decision-making. The story that is presented in *The Experience Machine* is internally consistent and elegant, offering nothing less than a comprehensive world view of how the brain and mind might work. But who knows what prediction errors science will throw up next.

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Competing interests

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